

Orthogonal Frequency Division Multiplexing (OFDM)

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Wideband Wireless Channels

- ▶ Wideband channel tends to
 - ▶ be frequency-selective since transmission bandwidth is typically more than channel coherence bandwidth
 - ▶ have large delay spread compared to symbol duration
- ▶ Wideband channel can be modeled using L number of taps:

$$y[m] = \sum_{l=0}^{L-1} h_l[m]x[m-l] + w[m] \quad (1)$$

- ▶ There is **Inter-Symbol Interference (ISI)**
- ▶ **Detection in wideband channel has to address ISI** (directly or indirectly)

Wideband Wireless Channels - Detection Approaches

- ▶ Three detection approaches for wideband channels
 - ▶ Single-carrier systems with equalization
 - ▶ Direct-sequence spread-spectrum
 - ▶ Multi-carrier systems
 - ▶ Example: OFDM
 - ▶ OFDM is used in major cellular standards like 4G/LTE and 5G/NR.

OFDM: Introduction

- ▶ Wideband channel model: $y[m] = \sum_{l=0}^{L-1} h_l[m]x[m-l] + w[m]$
- ▶ Rough outline of OFDM (ignoring details, which are covered later)
 - ▶ Let us ignore m dependence of $h_l[m]$
 - ▶ $y[m] = \sum_{l=0}^{L-1} h_l[m]x[m-l] + w[m]$
 - ▶ Frequency domain view (DFT turns convolution into a product)
 - ▶ $\tilde{y}_n = \tilde{h}_n \tilde{x}_n + \tilde{w}_n$ for each subcarrier n
 - ▶ Looks similar to channels with diversity we saw previously with following differences:
 - ▶ Data sent in frequency domain using $\tilde{x}_n$, instead of $x[m]$
 - ▶ Process DFT of received signal $\tilde{y}_n$, instead of the received signal $y[m]$
- ▶ OFDM coverts wideband channel with ISI into orthogonal narrowband subcarriers

OFDM: Setting

- ▶ Consider frequency selective channel with finite number L of taps: $y[m] = \sum_{l=0}^{L-1} h_l[m]x[m-l] + w[m]$
 - ▶ Information is sent in blocks of size N_c (N_c is block size)
 - ▶ We assume that h_l does not depend on m , i.e., N_c/W is much smaller than coherence time T_c . W is bandwidth of the wideband channel.
- ▶ Transmit $\mathbf{x}$ comprising $N_c + L - 1$ symbols of the following form: $\mathbf{x} = [d[N_c - L + 1], d[N_c - L + 2], \dots, d[N_c - 1], d[0], d[1], \dots, d[N_c - 1]]^T$
 - ▶ The red part above is cyclic prefix. Will soon see role of cyclic prefix.

OFDM basics: ISI-handling

- ▶ Consider the received signal, $y[m] = \sum_{l=1}^{L-1} h_l x[m-l] + w[m]$,
 $m = 1, 2, \dots, N_c + L - 1$
- ▶ Receiver ignores the first $L - 1$ samples to remove ISI from symbols related to previous block and just considers:
 $\mathbf{y} = [y[L], \dots, y[N_c + L - 1]]^T$
- ▶ **Efficiency loss:** $\frac{L-1}{N_c+L-1}$ fraction of time is “lost”
 - ▶ Large L and small N_c increases the loss

OFDM basics: Take DFT of received signal

- ▶ Taking N_c -point DFT of $\mathbf{y}$, we have,

$$\tilde{y}_n = \tilde{h}_n \tilde{d}_n + \tilde{w}_n, n = 0, 1, \dots, N_c - 1 \quad (2)$$

- ▶ where for $n = 0, 1, \dots, N_c - 1$, $\tilde{h}_n$ and $\tilde{d}_n$ are defined later, and

$$\tilde{y}_n = \frac{1}{\sqrt{N_c}} \sum_{m=0}^{N_c-1} y[m+L] \exp\left(\frac{-j2\pi nm}{N_c}\right) \quad (3)$$

$$\tilde{w}_n = \frac{1}{\sqrt{N_c}} \sum_{m=0}^{N_c-1} w[m+L] \exp\left(\frac{-j2\pi nm}{N_c}\right) \quad (4)$$

OFDM basics: DFT of convolution

- ▶ Considering right hand side of $\tilde{y}_n$, multiplied by $\sqrt{N_c}$

$$\begin{aligned} & \sum_{m=0}^{N_c-1} \left(\sum_{l=0}^{L-1} h_l x[m+L-l] \right) \exp \left(\frac{-j2\pi nm}{N_c} \right) \\ &= \sum_{l=0}^{L-1} h_l \exp \left(\frac{-j2\pi ln}{N_c} \right) \left(\sum_{m=0}^{N_c-1} x[m+L-l] \exp \left(\frac{-j2\pi n(m-l)}{N_c} \right) \right) \end{aligned}$$

- ▶ The term in the large brackets above can be written as,

$$\begin{aligned} & \sum_{m=0}^{N_c-1} x[m+L-l] \exp \left(\frac{-j2\pi n(m-l)}{N_c} \right) \\ &= \sum_{m'=-l}^{N_c-l-1} x[m'+L] \exp \left(\frac{-j2\pi nm'}{N_c} \right) \\ &= \sum_{m'=0}^{N_c-l-1} x[m'+L] \exp \left(\frac{-j2\pi nm'}{N_c} \right) + \sum_{m'=-l}^{-1} x[m'+L] \exp \left(\frac{-j2\pi nm'}{N_c} \right) \end{aligned}$$

OFDM basics: Role of cyclic prefix

- ▶ Since $\mathbf{x}$ starts with cyclic prefix, we have,

$$\begin{aligned} & \sum_{m'=0}^{N_c-L-1} x[m' + L] \exp\left(\frac{-j2\pi n m'}{N_c}\right) + \sum_{m'=-L}^{-1} x[m' + L] \exp\left(\frac{-j2\pi n m'}{N_c}\right) \\ &= \sum_{m'=0}^{N_c-L-1} d[m'] \exp\left(\frac{-j2\pi n m'}{N_c}\right) + \sum_{m'=-L}^{-1} d[N_c + m'] \exp\left(\frac{-j2\pi n m'}{N_c}\right) \\ &= \sum_{m'=0}^{N_c-L-1} d[m'] \exp\left(\frac{-j2\pi n m'}{N_c}\right) + \sum_{m'=-L}^{-1} d[N_c + m'] \exp\left(\frac{-j2\pi n (N_c + m')}{N_c}\right) \\ &= \sum_{m'=0}^{N_c-1} d[m'] \exp\left(\frac{-j2\pi n m'}{N_c}\right) = \tilde{d}_n \end{aligned}$$

- ▶ Roles of cyclic prefix
 - ▶ Some prefix of at least $L - 1$ redundant symbols needed to remove ISI
 - ▶ Cyclic prefix also decomposes the channel into orthogonal subcarriers

OFDM basics: In frequency domain

- ▶ Taking N_c – point DFT of CP-OFDM signal $\mathbf{y}$ (after removing CP) can convert wideband channel into N_c parallel subcarriers,

$$\tilde{y}_n = \tilde{h}_n \tilde{x}_n + \tilde{w}_n, n = 0, 1, \dots, N_c - 1$$

where,

$$\begin{aligned}\tilde{y}_n &= \frac{1}{\sqrt{N}} \sum_{m=0}^{N_c-1} y[m+L] \exp\left(\frac{-j2\pi nm}{N_c}\right) \\ \tilde{w}_n &= \frac{1}{\sqrt{N}} \sum_{m=0}^{N_c-1} w[m+L] \exp\left(\frac{-j2\pi nm}{N_c}\right) \\ \tilde{x}_n &= \frac{1}{\sqrt{N}} \sum_{m=0}^{N_c-1} x[m+L] \exp\left(\frac{-j2\pi nm}{N_c}\right) \\ \tilde{h}_n &= \sum_{l=0}^{L-1} h_l \exp\left(\frac{-j2\pi ln}{N_c}\right)\end{aligned}$$

OFDM basic: Subcarriers

- ▶ Consider the N_c subcarriers where,

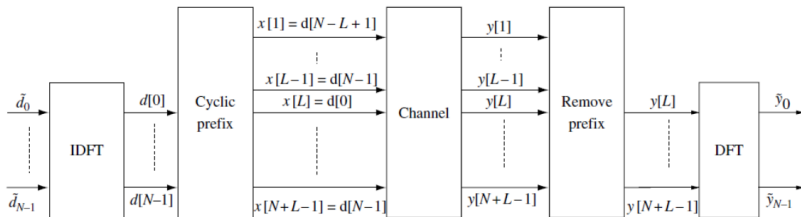
$$\tilde{y}_n = \tilde{h}_n \tilde{d}_n + \tilde{w}_n, n = 0, 1, \dots, N_c - 1$$

$$\text{where } \tilde{h}_n = \sum_{l=0}^{L-1} h_l \exp\left(\frac{-jl2\pi n}{N_c}\right)$$

- ▶ If OFDM is used in a wideband channel with bandwidth W , then **two adjacent subcarriers are separated by W/N_c**
 - ▶ Since, W -normalized angular frequency of subcarriers n and $n+1$ is separated by $\frac{2\pi}{N_c}$

OFDM: Transmission and reception

- ▶ OFDM operation can be viewed as,
 - ▶ Block of N_c data symbols $\tilde{d}_n, n = 0, 1, \dots, N_c - 1$ are in frequency domain
 - ▶ Transmit $x[m], m = 1, \dots, N_c + L - 1$, using **inverse DFT** of data symbols $(d_n)_n$ sent over the channel after adding cyclic prefix
 - ▶ Received signal is $y[m], m = 1, \dots, N_c + L - 1$
 - ▶ **Remove prefix-related symbols** from received signal, i.e., first $L - 1$ symbols
 - ▶ **Detection based on DFT** $\tilde{y}_n, n = 0, 1, \dots, N_c - 1$ of received signal. For subcarrier n , $\tilde{y}_n = \tilde{h}_n \tilde{d}_n + \tilde{w}_n$



OFDM: Detection

- ▶ The following model resembles received signal with multiple signal paths for narrowband channels

$$\tilde{y}_n = \tilde{h}_n \tilde{d}_n + \tilde{w}_n, n = 0, 1, \dots, N_C - 1$$

- ▶ It can be shown that $\tilde{w}_n, n = L, \dots, N_C + L - 1$, has same distribution as $w[L], n = 0, 1, \dots, N_C - 1$
- ▶ However, adjacent subcarriers can be correlated, and consequently $\tilde{h}_n$ may not be independent of $\tilde{h}_{n+1}$
- ▶ This is because adjacent subcarriers are separated by W/N_C , where as coherence bandwidth $W_C \approx \frac{1}{2T_d} = \frac{W}{2L}$ of the channel can be larger

OFDM performance: Diversity

- ▶ If subcarriers are sufficiently apart, we can obtain diversity gain
 - ▶ Because we can approximately treat $\tilde{h}_n$ for such sufficiently-apart subcarriers as independent
- ▶ How to ensure that subcarriers are sufficiently apart?
 - ▶ Adjacent subcarriers are separated by W/N_c
 - ▶ Coherence bandwidth of the channel: $W_c \approx \frac{1}{2T_d} = \frac{W}{2L}$
 - ▶ **Ideal interleaving in frequency: For any user,** if use subcarriers that are at least $2 \left(\frac{W}{2L}\right) = \frac{W}{L}$ apart, we can approximately treat $\tilde{h}_n$ for associated L subcarriers as independent (**Note that there are $L = \frac{W}{W/L}$ such subcarriers for each user**)
 - ▶ **With ideal interleaving in frequency, diversity gain up to L can be obtained by utilizing detection schemes (e.g., MRC) with $\tilde{y}_n = \tilde{h}_n \tilde{d}_n + \tilde{w}_n$, $n \in L$ indices of subcarriers separated by W/L**

OFDM performance: Block size N_c and high PAPR

- ▶ Choosing block size N_c
 - ▶ Due to cyclic prefix, $(L - 1)/(N_c + L - 1)$ fraction of symbols are not utilized and associated power is also wasted. Then, **why not use very high N_c ?**
 - ▶ N_c choice is limited by coherence time T_c of the channel
 - ▶ We assumed that channel does not change with time within duration of transmission of N_c symbols
 - ▶ Thus, N_c choice is driven by $\frac{N_c}{W} \ll T_c$
- ▶ OFDM's transmitted signal has **high peak-to-average-power ratio (PAPR)**
 - ▶ Peak power impacts biasing of RF components
 - ▶ High peak power leads to power-inefficient biasing
 - ▶ Thus, a significant consideration for (uplink of) power-constrained devices